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Catalytic upgrading of ethanol to higher alcohols over ordered mesoporous Cu-La-Al composite metal oxides

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ABSTRACT

Ordered mesoporous Cu-La-Al composite metal oxides were first synthesized via an evaporation induced self-assembly method and applied in upgrading of ethanol to higher alcohols. The Cu-La-Al composite metal oxide with optimal molar ratio of Cu/La/Al (4/2.6/100) exhibited selectivity and yield to higher alcohols up to 72.2% and 37.7%, respectively. What's more remarkable was that the catalyst exhibited ultra-high stability during 500 h on-stream evaluation. Various characterization techniques including N_2 adsorption-desorption, XRD, TEM, TPR, XPS, UV-vis, ^{27}Al NMR, NH_3 -TPD and CO_2 -TPD revealed that the synergy of a large amount of octahedrally coordinated Cu^{2+} and acid-base sites on the surface of Cu-La-Al composite metal oxides was responsible for the excellent catalytic activity and selectivity while the superior stability of the catalyst could be largely attributed to presence of the highly stable CuAl_2O_4 active phase besides the embedding effect of CuAl_2O_4 and La_2O_3 into skeleton of the composite oxides.

1. Introduction

Ethanol from sugar fermentation is the most productive biofuel in renewable resources [1]. USA, Brazil, PR China, and some European countries currently encourage the use of ethanol blended gasoline politically [2]. However, the ethanol content in the ethanol blended gasoline is limited to less than 20%, in order to avoid influencing the fuel quality and damaging the internal combustion engine [3]. Compared with ethanol, n-butanol has the following advantages: higher energy density ($29.2 \text{ MJ}\cdot\text{L}^{-1}$ vs. $19.6 \text{ MJ}\cdot\text{L}^{-1}$), lower corrosivity and more suitable for existing pipeline transportation; moreover, n-butanol is not only considered as an ideal substitute for gasoline, but also as the fuel of compression ignition engine without changing the engine hardware [4,5]. In addition, n-butanol can be used to synthesize butyl acrylate, butyl acetate, dibutyl phthalate and other important chemicals, which are widely used in plastic, rubber, paint, medicine and special solvent industries [6].

Catalytic coupling of ethanol to n-butanol is realized by the Guerbet mechanism (Scheme 1) [7,8], which consists of four cascaded steps: ethanol dehydrogenation, aldol condensation of acetaldehyde, dehydration of 3-hydroxybutyraldehyde, and hydrogenation of crotonaldehyde. Then the n-butanol can react with ethanol or itself in a similar way

to form 1-hexanol, 2-ethyl-1-butanol and other higher alcohols. In the last decades, both homo- and heterogeneous catalysts were widely explored for this reaction. Homogeneous research has focused on Iridium and Ruthenium complexes catalysts under relatively mild conditions [9–14], and much high selectivity to higher alcohols can be obtained. However, heterogeneous catalysts and processes are preferable for the conversion of ethanol to higher alcohols in view of the drawbacks of homogeneous catalysts such as discontinuity of production and potential environmental pollution concerning the use of soluble bases and organic solvents. There are mainly four types of heterogeneous catalysts explored for this reaction: metal oxides or mixed metal oxides [15–17], alkali metal-modified zeolites [18], hydroxyapatite (HAP) [19,20] and supported metal catalysts [21–25]. Incorporation of transition metals into the porous acid-base materials has been proven to significantly increase the yields of higher alcohols, mainly because the dehydrogenation/hydrogenation steps, which were accomplished previously with acid-base centers, are greatly improved [26,27]. For example, $\text{Ni}/\text{Al}_2\text{O}_3$ catalyst showed 25% ethanol conversion and 80% selectivity to n-butanol in a stirred tank reactor under relatively mild conditions [28]. Al_2O_3 -supported Ni, Cu or Co catalysts investigated by Riittonen et al. also showed similar catalytic performance [29]. However, due to the strong C-C cleavage ability of metal Ni, there were

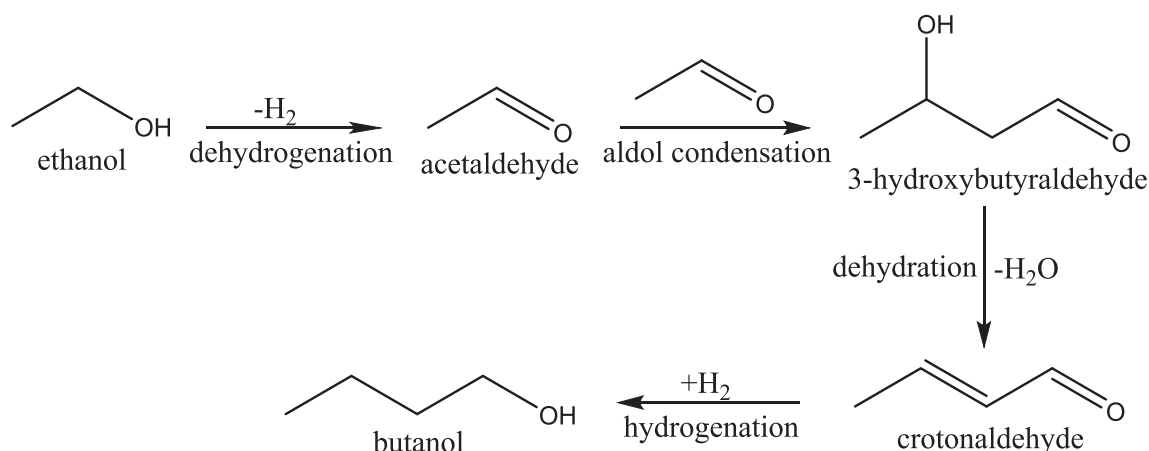
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Scheme 1. The Guerbet reaction pathway.

always CH_4 , CO , CO_2 and other cracking products detected in the tail gas [28]. Compared with the Ni based catalyst, the Cu based catalyst has higher ethanol dehydrogenation activity and zero cleavage capability of C-C bond under normal reaction conditions, therefore attracting more attention in recent years [27]. For instance, 80.2% selectivity to n-butanol and 95.5% total selectivity to linear alcohols at 33.5% conversion of ethanol were achieved with 5 %Cu/ Al_2O_3 catalyst in the supercritical regime (548 K, 10 MPa) [30]. Poliakoff and his co-workers [21] had exploited a high surface area CeO_2 supported Cu catalyst for upgrading ethanol to n-butanol in the presence of supercritical CO_2 , which exhibited 45% selectivity to and 30% yield of n-butanol, respectively. Also, We had demonstrated that Cu- CeO_2 /AC catalysts could exhibit ethanol conversion of 46.2% with selectivity to n-butanol of 41.3% and total selectivity to C4-C8 alcohols of 65% under relatively mild reaction conditions [22].

Although Cu based catalysts have not only high activity/selectivity but also low price, its inferior thermal stability caused by sintering of Cu nanoparticles are always a troublesome problem [21,22,31]. One generally used way to improve the thermal stability of Cu based catalysts is to load Cu on some supports with large surface area, such as AC [22,24], SiO_2 [32], Al_2O_3 [29,33] etc. Due to the stronger interaction between Cu and supports or promoters, the sintering resistance of Cu-based catalysts has been improved to a certain extent. Also, the stability of metal nanoparticles can be improved by coating SiO_2 porous membrane on the metal surface by chemical vapor deposition, preparing core-shell or metal semi-embedded structure catalysts, but the preparation processes of these catalysts are mostly complex, and it is usually difficult to improve the activity and stability of the catalysts at the same time [34–36]. Owing to high specific surface area, ordered mesoporous pore and higher thermal stability, ordered mesoporous alumina has been considered as an ideal catalyst support [37–39]. Additionally, Lanthanum-modified alumina support can provide the appropriate Lewis acid and base sites for the aldol condensation reaction, the elementary step in ethanol upgrading to n-butanol [33]. Therefore, in view of further improving the efficiency and especially stability of Cu-based catalysts for ethanol upgrading to n-butanol, ordered mesoporous Cu-La-Al composite metal oxide catalysts were first synthesized through an evaporation induced self-assembly method (EISA), and applied in upgrading of ethanol to higher alcohols. The structural properties of these catalysts were subsequently investigated to elucidate the reasons for their excellent catalytic performance.

2. Experimental

2.1. Catalyst preparation

All chemicals used in this procedure were AR grade. A series of

ordered mesoporous Cu-La-Al composite metal oxides were prepared by EISA method as follows: 2 g of P123 ($EO_{20}PO_{70}EO_{20}$) was firstly dissolved in 20 mL anhydrous ethanol. Then, 0.286 g salicylic acid, 0.2 g citric acid and calculated amounts of $La(Ac)_3 \cdot xH_2O$ and $Cu(Ac)_2$ were added into above solution. With complete dissolution of all solutes, a clear solution denoted solution A was achieved. Solution B was achieved by adding 3.2 mL 35 wt% HCl and subsequently 4.08 g (20 mmol) aluminum isopropoxide into another 20 mL anhydrous ethanol. After the solution A and B were mixed and stirred vigorously for 5 h at room temperature, the ethanol solvent was evaporated at 333 K for 72 h. The resulting solid samples were heated at a rate of 1 K/min to 673 K for 4 h, and then calcined at 973 K for 1 h with a heating rate of 10 K/min. The ordered mesoporous Cu-La-Al composite metal oxides obtained finally were denoted as OM-Cu_xLa_yAl₁₀₀, where the subscripts indicate the relative molar number (RMN) of Cu, La and Al, respectively. As reference, ordered mesoporous Al_2O_3 , La-Al and Cu-Al composite metal oxides (denoted as OMA, OM-Cu_xAl₁₀₀, and OM-La_yAl₁₀₀, respectively) were prepared in the same process without the addition of lanthanum acetate and/or copper acetate.

Stoichiometric copper aluminate spinel ($CuAl_2O_4$) catalyst was prepared by grinding $Al(NO_3)_3 \cdot 9H_2O$ and $Cu(NO_3)_2 \cdot 3H_2O$ with Al/Cu molar ratio of 2:1 to uniform powder mixture in quartz inkstone. The powder mixture was then calcined at 1073 K for 6 h with a heat ramp of 10 K/min and then the $CuAl_2O_4$ sample was obtained.

2.2. Catalyst characterization

Brunauer-Emmett-Teller (BET) surface areas of the catalysts were determined by N_2 adsorption using a Quantachrome ASIQC0 FV100•2 apparatus with liquid- N_2 at the temperature of 77 K. The samples were outgassed at 473 K for 4 h prior to analysis. X-ray diffraction (XRD) data were collected on an X'Pert PRO X-ray diffractometer employing a $Cu-K\alpha$ radiation source ($\lambda = 1.5406 \text{ \AA}$) and operated at 40 kV and 40 mA. The scanning rate was $0.008^\circ \text{ s}^{-1}$ in the 2θ range from 0.5° to 5° for small-angle XRD and $0.05^\circ \text{ s}^{-1}$ from 10° to 80° for wide-angle XRD, respectively. High resolution transmission electron microscopy (HRTEM) was performed using a Talos S-FEG operated at 200 kV from Philips-FEI Company. Subångström resolution high-angle annular dark field scanning transmission electron microscopy (HAADF STEM) images were obtained on a JEOL JEM 2200FS STEM/TEM, equipped with a CEOS (Heidelberg, Ger) probe corrector, and a nominal image resolution of 0.07 nm. Before microscopy examination, the samples were dispersed onto a molybdenum grid coated with a thin holey carbon film. X-ray photoelectron spectroscopy (XPS) spectra were obtained with an ESCALab220i-XL electron spectrometer from VG Scientific using 300 W $AlK\alpha$ radiation and deconvoluted using XPS PEAK41 software. The fresh catalyst was transferred directly to XPS chamber while the catalyst after

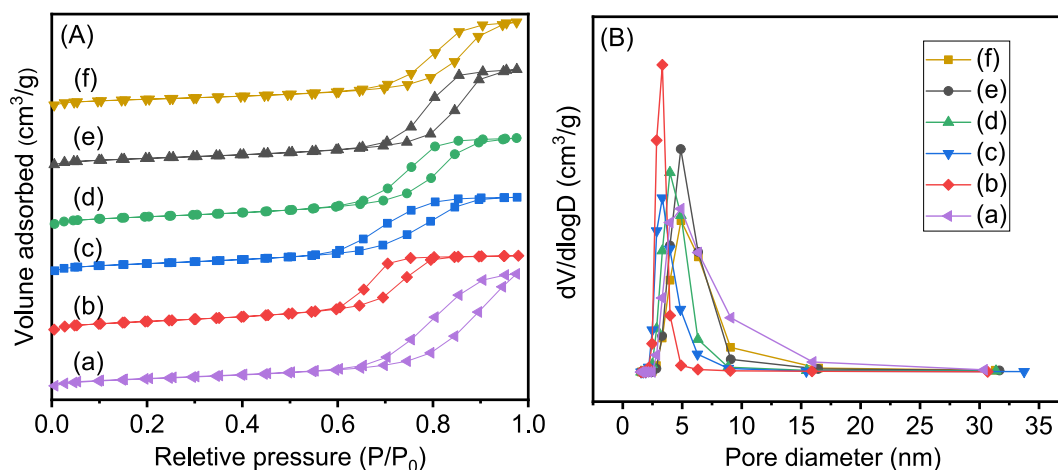


Fig. 1. N₂ adsorption-desorption isotherm (A) and pore size distribution curves (B) of (a) OMA, (b) OM-Cu₄Al₁₀₀, (c) OM-La_{2.6}Al₁₀₀, (d) OM-Cu_{0.8}La_{2.6}Al₁₀₀, (e) OM-Cu₄La_{2.6}Al₁₀₀, (f) OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalysts.

the reaction was transferred into the XPS chamber under Ar protection. The binding energies (BE) were referenced to the graphitic C 1s peak at 284.6 eV, providing accuracy within ± 0.2 eV. UV-vis spectra were collected with a Shimadzu UV-3600 instrument equipped with an integrating sphere. The samples were mounted in quartz cell recorded in the range 200–2000 nm at room temperature and BaSO₄ was used as reference. ²⁷Al magic angle spinning (MAS) NMR spectra were recorded on an Agilent 600 DD2 spectrometer operating at a Larmor frequency of 156.25 MHz for ²⁷Al.

H₂ temperature-programmed reduction (H₂-TPR) were performed on a Micromeritics Autochem II 2920 chemisorber equipped with a thermal conductivity detector (TCD) and a OmniStar GSD320 mass spectrometer (MS). Typically, the sample (100 mg) was pretreated in a quartz U-tube reactor with a He gas stream (30 mL/min) at 473 K for 1 h. After cooling to room temperature, a 10% H₂/Ar flow (50 mL/min) was introduced and the sample was heated to 1273 K (10 K/min) while desorbed species were monitored by the thermal conductivity detector (TCD) and mass spectrometry (MS) in real time.

CO₂ temperature-programmed desorption (CO₂-TPD) and NH₃ temperature-programmed desorption (NH₃-TPD) were also performed on the Micromeritics Autochem II 2920 chemisorber. For CO₂-TPD experiments, the sample (100 mg) was pretreated in a quartz U-tube reactor with a He gas stream (30 mL/min) at 473 K for 1 h. When the temperature decreased to 323 K, CO₂ saturation uptake of the pretreated catalysts was achieved by passing pure CO₂ for 0.5 h at 30 mL/min. After the CO₂ adsorption, the catalyst was purged by Ar flow (40 mL/min) for 1 h at 323 K. Then the temperature was linearly increased from 323 K to 1073 K (10 K/min) in Ar flow while desorbed species were monitored by the thermal conductivity detector (TCD) and mass spectrometry (MS) in real time. NH₃-TPD experiments were carried out by similar procedure as that of CO₂-TPD experiments except for using 10% NH₃/Ar gas mixture as the adsorbate.

2.3. Catalyst evaluation

The catalytic tests were performed in a fixed-bed reactor as shown in our previous paper [25]. The reaction was carried out under the following conditions: 533 K, 3 MPa, LHSV = 2 mL/(h·g_{cat}) and N₂/ethanol = 250:1 (v/v). The liquid products were analyzed by a Gas chromatograph (GC) with a flame ionization detector (FID) and an HP-5 column (30 m, 0.25 mm inner diameter) calibrated with analytes.

The ethanol conversion, selectivity and yield of products were calculated as follows:

$$\text{Ethanol conversion (\%)} = \frac{C \text{ mol of products}}{C \text{ mol of (products + unreacted ethanol)}} \times 100\%$$

$$\text{Product selectivity (\%)} = \frac{C \text{ mol of specific product}}{C \text{ mol of products}} \times 100\%$$

$$\text{Product yield (\%)} = \text{Ethanol conversion} \times \text{Product selectivity}$$

where C mol was the mole number of carbon in the products and unreacted ethanol.

The carbon balance for this reaction was calculated as follows:

$$\text{Carbon balance (\%)} = \frac{C \text{ mol of (products + unreacted ethanol)}}{C \text{ mol of inlet ethanol}} \times 100\%$$

where the meaning of C mol was the same as above.

The carbon balances calculated for ethanol upgrading to higher alcohols over the OM-Cu_xLa_yAl₁₀₀ catalysts were nearly 100%, which is expected in view of the extremely low cleavage capacity for C-C bonds of Cu-based catalysts [22,27].

3. Results and discussion

3.1. N₂ adsorption-desorption analysis

N₂ adsorption-desorption isotherm curves and pore-size distribution of this series catalysts were shown in Fig. 1. All the samples presented Type IV with H1-shaped hysteresis loops at higher relative pressure (P/P₀), suggesting the presence of a well-developed mesoporous structure for the newly synthesized catalysts [40]. There were steep condensation steps in the adsorption isotherm curves, implying very narrow pore-size distributions as displayed in Fig. 1(B). The textural parameters of these samples were summarized in the Table 1. The textural parameters of OMA are comparable to those of ordered mesoporous alumina reported in recent literature [41,42]. With introduction of copper into OMA catalyst, though the S_{BET} increased from 254.4 m²·g⁻¹ to 276.4 m²·g⁻¹, V_t and D_{pore} decreased from 1.06 cm³·g⁻¹ and 7.9 nm to 0.75 cm³·g⁻¹ and 6.7 nm, respectively. All of the S_{BET}, V_t and D_{pore} decreased with addition of lanthanum into OMA catalyst. For the series of catalysts containing lanthanum, as the RMN of Cu increased from 0 to 6.3, the S_{BET} and V_t initially increased and then decreased with the maximum of 254.5 m²·g⁻¹ and 0.91 cm³·g⁻¹ at the Cu RMN of 0.8 and 4, respectively, while the D_{pore} showed a gradually increasing trend. The S_{BET} and V_t

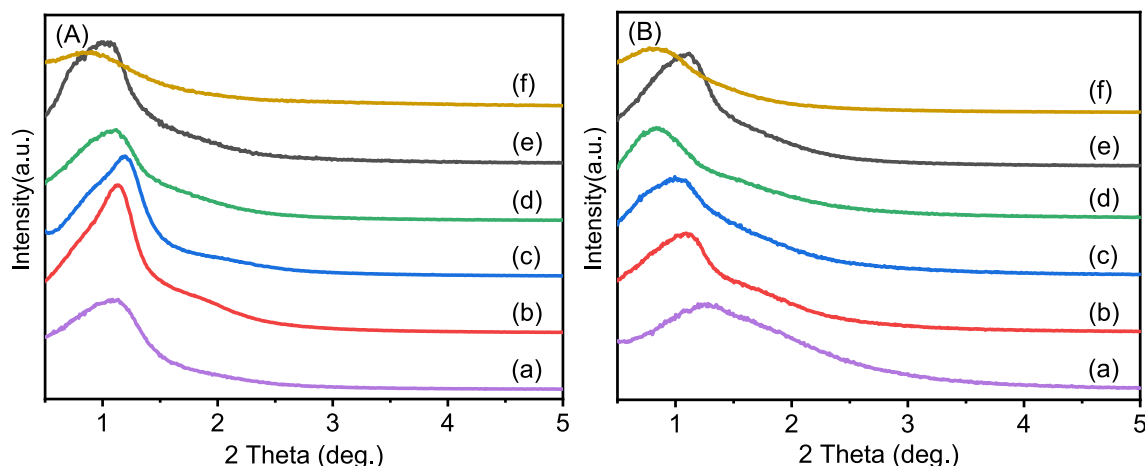


Fig. 2. Small-angle XRD spectra of (a) OMA, (b) OM-Cu₄Al₁₀₀, (c) OM-La_{2.6}Al₁₀₀, (d) OM-Cu_{0.8}La_{2.6}Al₁₀₀, (e) OM-Cu₄La_{2.6}Al₁₀₀, and (f) OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalysts before reaction (A) and after 12 h reaction (B).

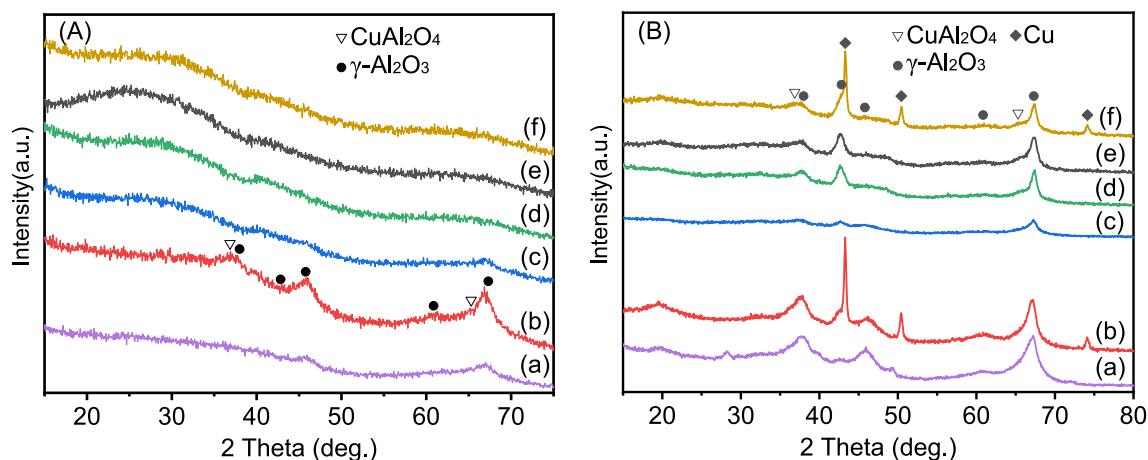


Fig. 3. Wide-angle XRD spectra of (a) OMA, (b) OM-Cu₄Al₁₀₀, (c) OM-La_{2.6}Al₁₀₀, (d) OM-Cu_{0.8}La_{2.6}Al₁₀₀, (e) OM-Cu₄La_{2.6}Al₁₀₀, and (f) OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalysts before reaction (A) and after 12 h reaction (B).

obviously decreased with excessive increase of the Cu RMN (6.3). The high specific surface area, large pore volume and ordered mesoporous pore of the Cu_xLa_yAl₁₀₀ composite oxides provide a premise for them to be used as excellent catalysts.

3.2. XRD and TEM analysis

Small-angle XRD patterns of this series of catalysts before and after 12 h reaction were showed in the Fig. 2. Except for OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalyst, all catalysts displayed a strong diffraction peak at around 1° and a weak shoulder peak at around 1.5°, which are respectively characteristic of (100) and (110)/(200) planes of ordered mesoporous structure with a 2D hexagonal *p6mm* symmetry [43]. A weak diffraction peak at around 1° was observed for the OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalyst before reaction, indicating its low structure order, which is consistent with the results of N₂ adsorption. Each catalyst after the reaction displayed almost the same diffraction spectrum as that of the corresponding catalyst before reaction. This result suggests the ordered mesoporous structure of catalysts was well preserved during ethanol upgrading to higher alcohols.

Wide-angle XRD patterns of this series of catalysts before and after 12 h reaction were showed in the Fig. 3. Newly synthesized OMA exhibited two weak diffraction peaks characteristic of γ -Al₂O₃ [44], suggesting it is essentially γ -Al₂O₃ with lower degree of crystallization.

For the OM-Cu₄Al₁₀₀ catalyst before reaction, very weak diffraction peaks characteristic of CuAl₂O₄ crystal phase were found [45], while intensities of diffraction peaks characteristic of γ -Al₂O₃ increased, implying introduction of copper improved crystallization degree of the ordered mesoporous framework. However, intensities of the diffraction peaks characteristic of γ -Al₂O₃ decreased or disappeared with the incorporation of lanthanum; moreover, there were no diffraction peaks of Cu and La species observed for the catalysts containing lanthanum, indicating Cu and La species were highly dispersed or existed in amorphous forms in these catalysts.

As shown in Fig. 3 (B), all the catalysts after reaction showed clear diffraction peaks characteristic of γ -Al₂O₃, indicating the ordered mesoporous frameworks had undergone a gradual crystallization process during ethanol upgrading to higher alcohols. Sharp diffraction peaks at $2\theta = 43.3^\circ$, 50.4° and 74.1° characteristic of Cu nanoparticles were found while the diffraction peaks characteristic of CuAl₂O₄ found before reaction were weakened for the OM-Cu₄Al₁₀₀ catalyst after reaction. These results may be interpreted as reduction of a little active CuAl₂O₄ into metal Cu nanoparticles. However, there were no diffraction peaks characteristic of Cu nanoparticles but very diffuse diffraction peaks characteristic of CuAl₂O₄ observed for the OM-Cu_{0.8}La_{2.6}Al₁₀₀ and OM-Cu₄La_{2.6}Al₁₀₀ catalysts, suggesting the introduction of lanthanum prevented reduction of CuAl₂O₄ species in these catalysts or the CuAl₂O₄ in these catalysts probably existed in different physical and chemical

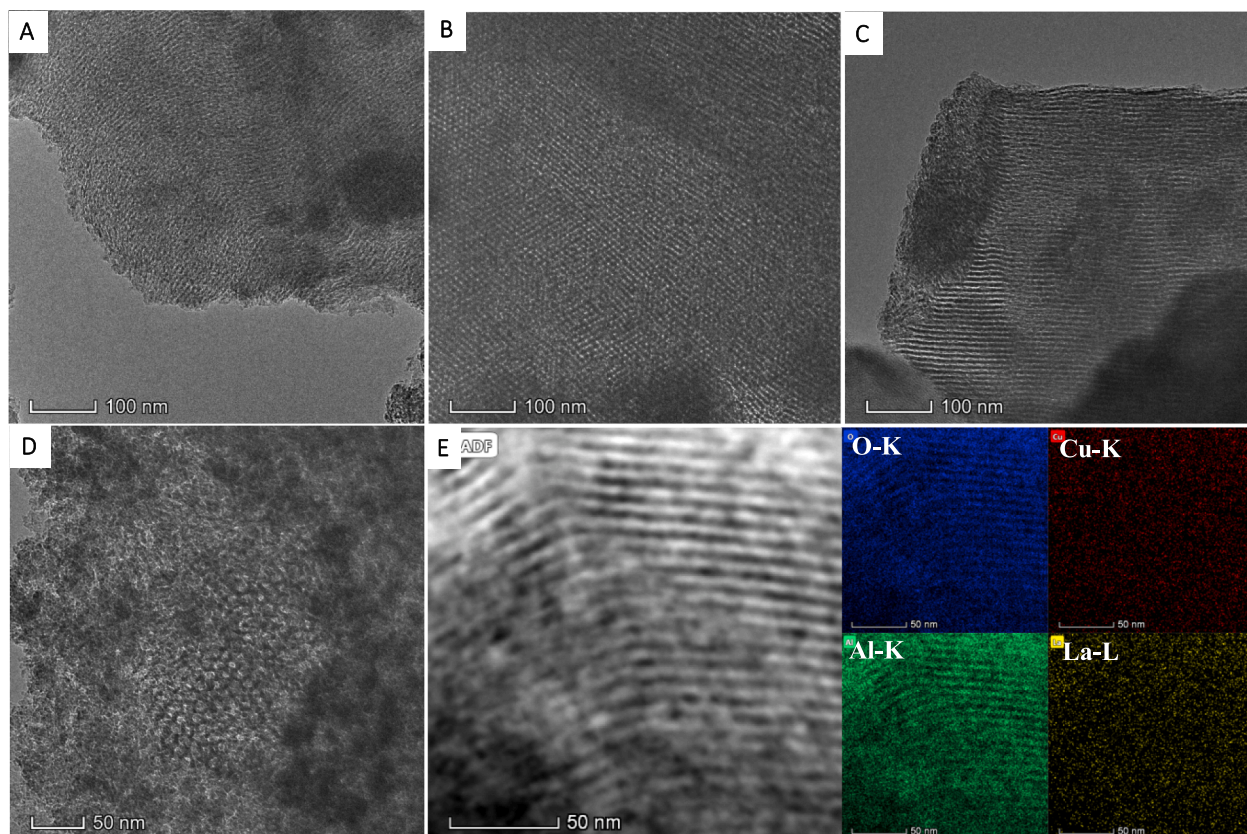


Fig. 4. TEM images of OM-Cu₄Al₁₀₀ (A), OM-Cu_{0.8}La_{2.6}Al₁₀₀ (B), OM-Cu₄La_{2.6}Al₁₀₀ (C), and OM-Cu_{6.3}La_{2.6}Al₁₀₀ (D) catalysts after reaction; HAADF-TEM images of OM-Cu₄La_{2.6}Al₁₀₀ catalyst after reaction and corresponding EDX mapping images (E).

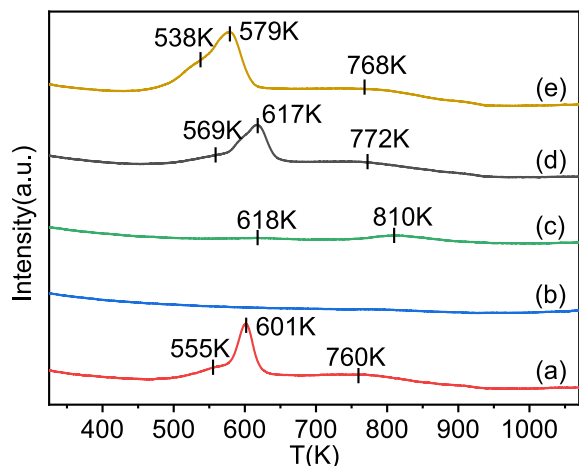


Fig. 5. H₂-TPR profiles of (a) OM-Cu₄Al₁₀₀, (b) OM-La_{2.6}Al₁₀₀, (c) OM-Cu_{0.8}La_{2.6}Al₁₀₀, (d) OM-Cu₄La_{2.6}Al₁₀₀, and (e) OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalysts.

environments. For the OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalyst after reaction, diffraction peaks characteristic of both metal Cu and CuAl₂O₄ were observed, due to diversification of Cu species with excessive introduction of copper. It is remarkable that the diffraction peaks characteristic of CuAl₂O₄ for the OM-Cu_xLa_{2.6}Al₁₀₀ catalysts ($x = 0.8, 2.6$ and 4) after reaction were very weak and diffuse, indicating the high dispersion of the CuAl₂O₄ species over the OM-Cu_xLa_{2.6}Al₁₀₀ catalysts. Additionally, due to the large size of La³⁺ compared to Al³⁺ (0.054 nm for Al³⁺ and 0.103 nm for La³⁺), it is difficult to form LaAlO₃ phase at present calcination temperature [46]. Thus, the lanthanum species were expected to exist in the form of highly dispersed La₂O₃ in these catalysts.

As seen in Fig. 4(A)–(D), hexagonally packed honeycomb-like mesopores and parallel cylindrical mesochannels along the (110) and (100) orientations could be clearly observed for OM-Cu₄Al₁₀₀, OM-Cu_{0.8}La_{2.6}Al₁₀₀ and OM-Cu₄La_{2.6}Al₁₀₀ catalysts though the ordered mesoporous framework locally collapsed for the OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalyst [47]. These observations are consistent with the former results of N₂ adsorption-desorption and Small-angle XRD. HAADF-TEM and EDX mapping images of the OM-Cu₄La_{2.6}Al₁₀₀ catalyst (Fig. 4(E)) showed that Al-K and O-K signals could only be detected in the ordered skeleton, further confirming that the ordered mesoporous framework was based on the alumina matrix; moreover, Cu-K and La-L signals were evenly spread over the selected area, implying the high dispersion of copper and lanthanum species, which is consistent with the former results of Wide-angle XRD.

3.3. H₂-TPR and XPS analysis

H₂-TPR profiles of this series of catalysts were shown in Fig. 5. There were three H₂ consumption peaks centered at 555 K, 601 K and 760 K, respectively, in the TPR curve of OM-Cu₄Al₁₀₀ catalyst. First reduction peak can be attributed to the reduction of the extra-framework or free CuAl₂O₄ (denoted as CuAl₂O₄-I) while the latter two peaks correspond to the stepwise reduction of the CuAl₂O₄ embedded on the ordered mesoporous framework (denoted as CuAl₂O₄-II) [48]. As shown in Fig. 5 (B), there was no H₂ consumption peak observed for the OM-La_{2.6}Al₁₀₀ catalyst, indicating strong reduction resistances of La₂O₃ and Al₂O₃. Comparing TPR curve of OM-Cu₄La_{2.6}Al₁₀₀ catalyst with that of OM-Cu₄Al₁₀₀ catalyst, it can be found that reduction temperature of three kinds of Cu species increased with the introduction of La, indicating a stronger interaction between lanthanum oxide and CuAl₂O₄ species than that between aluminum oxide and CuAl₂O₄ species. Additionally, the reduction temperature of three kinds of Cu species decreased with the

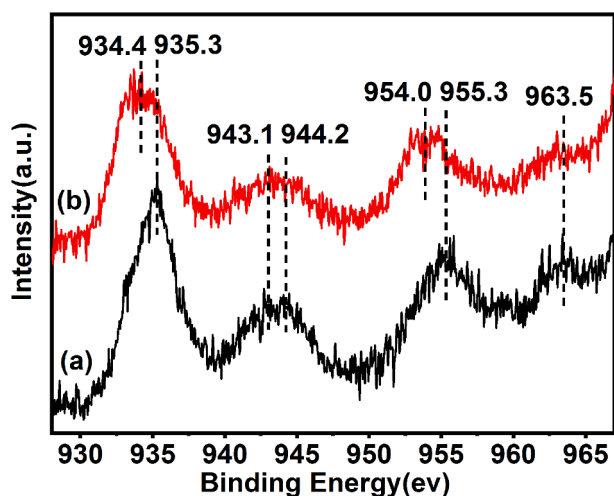


Fig. 6. Cu 2p XPS spectra of OM-Cu₄La_{2.6}Al₁₀₀ catalyst before (a) and after reaction (b).

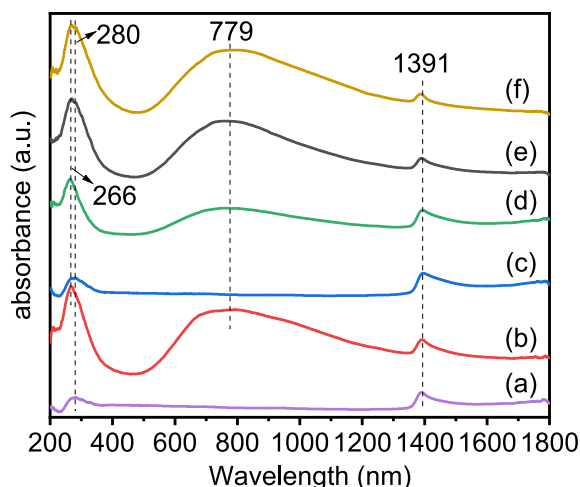


Fig. 7. UV-vis spectra of (a) OMA, (b) OM-Cu₄Al₁₀₀, (c) OM-La_{2.6}Al₁₀₀, (d) OM-Cu_{0.8}La_{2.6}Al₁₀₀, (e) OM-Cu₄La_{2.6}Al₁₀₀, and (f) OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalysts.

increase of Cu loading, which could be attributed to gradually decreased dispersion of CuAl₂O₄ species. Notably, the reduction peak of CuAl₂O₄-II was much stronger than that of CuAl₂O₄-I for the OM-Cu₄La_{2.6}Al₁₀₀ catalyst, indicating that the copper species on the surface of the OM-Cu₄La_{2.6}Al₁₀₀ catalyst existed mainly as CuAl₂O₄-II, namely the CuAl₂O₄ species embedded on the ordered mesoporous skeleton predominated over the OM-Cu₄La_{2.6}Al₁₀₀. Furthermore, since the initial temperature of this peak was about 580 K, which is higher than the reaction temperature (533 K), the most surface CuAl₂O₄ species could be difficult to reduce during the reaction of ethanol catalytic upgrading.

In order to further identify kinds and changes of the Cu species over OM-Cu₄La_{2.6}Al₁₀₀ catalyst, XPS characterization of OM-Cu₄La_{2.6}Al₁₀₀ catalyst before and after the reaction were performed. As shown in Fig. 6, presence of the peaks at 935.3 and 955.3 eV or at 934.4 and 954.0 eV accompanied by the satellite peaks at 940–947 eV indicated that the copper species on the surface of the catalysts both before and after the reaction existed in the form of Cu²⁺ [49,50]; moreover, the Cu 2P_{3/2} binding energies of OM-Cu₄La_{2.6}Al₁₀₀ catalysts both before and after the reaction were higher than that of CuO (935.3 and 934.4 eV vs. 933.7 eV), further demonstrating the existence of CuAl₂O₄ species [51,52]. Additionally, the Cu 2P_{3/2} binding energy of the catalyst after the reaction was slightly reduced, which may be related to the slight change in interaction among the CuAl₂O₄, La₂O₃ and alumina matrix

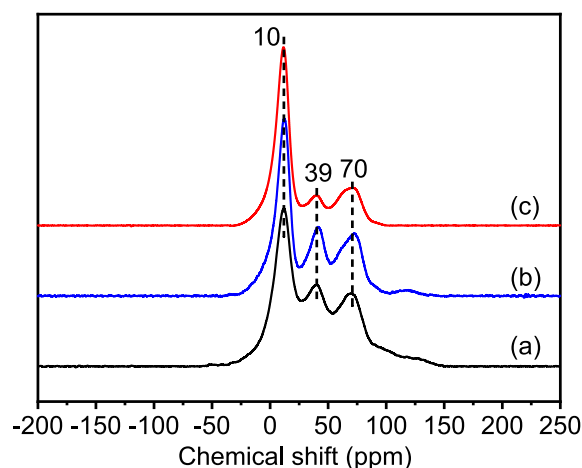


Fig. 8. Solid-state ²⁷Al NMR profiles of (a) OMA, (b) OM-La_{2.6}Al₁₀₀, and (c) OM-Cu₄La_{2.6}Al₁₀₀ catalysts.

during the ethanol upgrading. However, the most surface CuAl₂O₄ species were not reduced during the catalytic upgrading of ethanol due to the high binding energies and existence of the satellite peaks at 940–947 eV.

3.4. UV-vis and Solid-state ²⁷Al NMR analysis

UV-vis spectra of this series of catalysts were shown in Fig. 7. There were two adsorption bands at 280 nm and 1391 nm observed for the OMA sample. The former is assignable to electronic transition of Al³⁺ while the later can be assigned to the unresolved first overtone of surface hydroxyl groups [53,54]. After introduction of copper, a strong absorption band centered at 779 nm appeared accompanied by formation of an absorption peak at 266 nm; meanwhile the earlier adsorption peak at 280 nm changed into a shoulder peak of the new peak at 266 nm and the peak at 1391 nm was almost the same. The two adsorption bands at 266 and 779 nm are typically characteristic of octahedrally coordinated Cu²⁺ ions (denoted as Cu_{Octa}²⁺) in aluminate spinel, which is consistent with the results of Wide-angle XRD, TPR and XPS that the existence of a large amount of CuAl₂O₄ species consisting of both Cu_{Octa}²⁺ and tetrahedrally coordinated Cu²⁺ ions (denoted as Cu_{Tetra}²⁺) on the surface of OM-Cu₄Al₁₀₀ catalyst [54,55]. It is remarkable that for the OM-Cu₄Al₁₀₀ catalyst, there were no adsorption peaks at 475 nm and 1450 nm characteristic of Cu_{Tetra}²⁺ in aluminate spinel [56,57], suggesting the Cu²⁺ ions were predominantly in the octahedral sites. Previously, Cu²⁺ ions were found to preferentially occupy vacant octahedral sites on the surface of Al₂O₃, followed by allocation at the tetrahedral sites in a little deep layer of the crystallite structure; moreover, at a low Cu loading or surface Cu concentration, Cu²⁺ ions were predominantly in the octahedral sites [47,48,53]. This would explain the present result that the Cu_{Octa}²⁺ predominantly existed on the surface of OM-Cu_xLa_yAl₁₀₀ catalysts in view of their lower Cu loading and the high dispersion of CuAl₂O₄ species. For the samples containing both copper and lanthanum, the intensities of bands centered at 266 nm and 779 nm gradually increased with the increase of Cu content, indicating the increasing amount of the CuAl₂O₄ species and Cu_{Octa}²⁺. Additionally, UV-vis spectrum of La-OMA sample was almost the same as that of OMA sample, which further indicated that La³⁺ ions could not enter into the lattice of the Al₂O₃ matrix with the formation of LaAlO₃ species.

The chemical environments of Al³⁺ ions in this series of catalysts were investigated by solid-state ²⁷Al NMR, and the results were displayed in Fig. 8. All catalysts exhibited three resonance peaks at 10 ppm, 39 and 70 ppm, respectively. The former is characteristic of the octahedrally coordinated Al³⁺ ions (denoted as Al_{Octa}³⁺) while the latter two resonances peaks are attributed to the penta-coordinated Al³⁺ (denoted

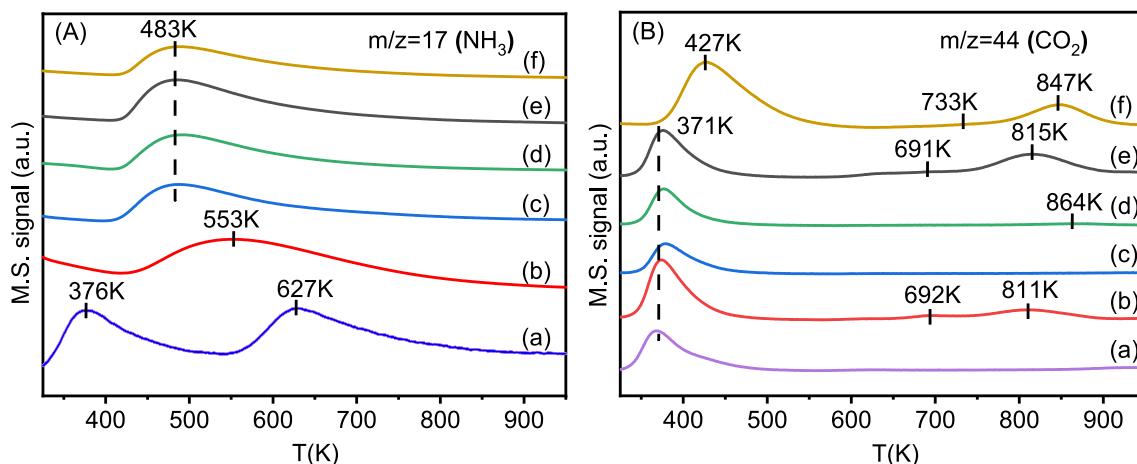


Fig. 9. NH_3 -TPD (A) and CO_2 -TPD (B) profiles of (a) OMA, (b) OM-Cu₄Al₁₀₀, (c) OM-La_{2.6}Al₁₀₀, (d) OM-Cu_{0.8}La_{2.6}Al₁₀₀, (e) OM-Cu₄La_{2.6}Al₁₀₀, and (f) OM-Cu_{6.3}La_{2.6}Al₁₀₀ catalysts.

Table 1

Textural parameters and surface properties of OMA and OM-Cu_xLa_yAl₁₀₀ catalysts.

Sample	$S_{\text{BET}}^{\text{[a]}}/\text{m}^2\cdot\text{g}^{-1}$	$V_{\text{t}}^{\text{[b]}}/\text{cm}^3\cdot\text{g}^{-1}$	$D_{\text{pore}}^{\text{[c]}}/\text{nm}$	$N_{\text{Oxide}}^{\text{[d]}}$	Total acidity/ $\text{mmol}\cdot\text{g}^{-1}$	Total basicity/ $\mu\text{mol}\cdot\text{g}^{-1}$	Basicity distribution ($\mu\text{mol}\cdot\text{g}^{-1}$)		
							Weak	Medium	Strong
OM-Cu _{6.3} La _{2.6} Al ₁₀₀	191.9	0.77	9.8	91.3	0.41	62.1	43.8	1.6	16.7
OM-Cu ₄ La _{2.6} Al ₁₀₀	231.8	0.91	9.7	77.0	0.68	43.4	19.3	2.8	21.3
OM-Cu _{0.8} La _{2.6} Al ₁₀₀	254.5	0.83	8.0	34.1	0.47	16.9	14.4	–	2.5
OM-La _{2.6} Al ₁₀₀	245.0	0.73	6.7	–	0.37	12.8	12.8	–	–
OM-Cu ₄ Al ₁₀₀	276.4	0.75	6.7	90.8	1.08	34.5	24.5	2.4	7.7
OMA	254.4	1.06	7.9	–	1.27	17.3	17.3	–	–

[a] Brunauer-Emmett-Teller surface area; [b] Single point pore volume at $P/P_0 = 0.99$; [c] The average pore diameters (DFT method); [d] The relative numbers of N_{Oxide}^{2+} calculated according to the areas of UV-vis adsorption peaks at 779 nm in Fig. 6.

as $\text{Al}_{\text{Penta}}^{3+}$ and tetrahedrally coordinated $\text{Al}_{\text{Tetra}}^{3+}$ ions (denoted as $\text{Al}_{\text{Tetra}}^{3+}$), respectively [58,59]. The ratio of $\text{Al}_{\text{Octa}}^{3+}/\text{Al}_{\text{Penta}}^{3+}/\text{Al}_{\text{Tetra}}^{3+}$ were quantitatively calculated as 4.7/2.1/3.2, 5.0/2.0/3 and 6.5/1.2/2.3 for OMA, OM-La_{2.6}Al₁₀₀ and OM-Cu₄La_{2.6}Al₁₀₀ catalysts respectively. For OM-Cu₄La_{2.6}Al₁₀₀ catalyst, a significant decrease of the amount of $\text{Al}_{\text{Penta}}^{3+}$ accompanied by obvious increase of the amount of $\text{Al}_{\text{Octa}}^{3+}$ suggested that Cu^{2+} ions preferentially coordinatively unsaturated $\text{Al}_{\text{Penta}}^{3+}$ to $\text{Al}_{\text{Octa}}^{3+}$ by the formation Cu-O-Al [60], which is consistent with the results of UV-vis. Also, a little Cu^{2+} ions were probably incorporated into the lattice of Al_2O_3 matrix by forming the $\text{Cu}_{\text{Tetra}}^{2+}$, resulting in a reduction of the amount of $\text{Al}_{\text{Tetra}}^{3+}$ [42]. Then, the condensation of the $\text{Cu}_{\text{Octa}}^{2+}$ and $\text{Cu}_{\text{Tetra}}^{2+}$ resulted in the formation of the highly dispersed CuAl_2O_4 species on the $\text{Cu}_x\text{La}_y\text{Al}_{100}$ [61]. Further, as confirmed by UV-vis results, the $\text{Cu}_{\text{Octa}}^{2+}$ predominantly existed on the surface of OM-Cu_xLa_yAl₁₀₀ catalysts, which can be interpreted by that Cu^{2+} ions preferentially occupy vacant octahedral sites on the surface of Al_2O_3 , followed by allocation at the tetrahedral sites in a little deep layer of the crystallite structure. Additionally, the ratio of $\text{Al}_{\text{Octa}}^{3+}/\text{Al}_{\text{Penta}}^{3+}/\text{Al}_{\text{Tetra}}^{3+}$ on the surface of OMA and OM-La_{2.6}Al₁₀₀ catalyst was almost the same, further indicating that La species had not bonded with the $\text{Al}_{\text{Penta}}^{3+}$ on Al_2O_3 matrix or formed LaAlO₃ species.

3.5. NH_3 -TPD and CO_2 -TPD analysis

Fig. 9(A) presented NH_3 -TPD profiles of this series of catalysts and total number and strength distribution of the acid sites were summarized in Table 1. As shown in Fig. 8(A-a), desorption peaks at 627 K and 376 K correspond to strong and weak acid sites, respectively, on the surface of OMA, which originate from the coordinatively unsaturated Al^{3+} ions [62] and -OH groups adjacent to them [63]. The introduction of copper reduced the number of the strong acid sites and almost eliminated the

weak acid sites, accompanied by formation of medium strength acid sites, which could be related to anchoring of Cu^{2+} ions at $\text{Al}_{\text{Penta}}^{3+}$ sites on surface of Al_2O_3 substrate by formation of Cu-O-Al bonds [64,65]. Similarly, there was a NH_3 desorption peak centered at 483 K corresponding to medium acid sites in the NH_3 -TPD curve of OM-La_{2.6}Al₁₀₀ catalyst, indicating the highly dispersed La₂O₃ neutralized the weak acid sites and decreased the strength of the strong acid on the surface of Al_2O_3 substrate [62]. Further addition of Cu on the basis of OM-La_{2.6}Al₁₀₀ catalyst almost did not change the strength of the moderate acid sites, but number of the moderate acid sites increased from 0.37 $\text{mmol}\cdot\text{g}^{-1}$ to 0.68 $\text{mmol}\cdot\text{g}^{-1}$ with increase of the RMN of Cu from 0 to 4 and then decreased to 0.41 $\text{mmol}\cdot\text{g}^{-1}$ probably due to the collapse of ordered mesoporous framework.

CO_2 -TPD profiles of this series of catalysts were shown in Fig. 9(B) and total number and strength distribution of the basic sites were summarized in Table 1. It can be observed that OMA exhibited a CO_2 desorption peak centered at 371 K, corresponding to weak basic sites due to the basic -OH groups on the surface of OMA [66]. The number of the weak basic sites increased with the introduction of copper, which could be attributed to increase of the basic -OH groups with anchoring of Cu^{2+} ions at $\text{Al}_{\text{Penta}}^{3+}$ sites by formation of Cu-O-Al bonds. Increase of S_{BET} from 254.4 m^2/g to 276.4 m^2/g is probably another explanation for the increase of the number of the weak basic sites. Additionally, a small number of medium and strong basic sites on the OM-Cu₄Al₁₀₀ catalyst are maybe related to the $\text{Cu}^{2+}-\text{O}^{2-}$ pairs and the coordinatively unsaturated oxygen anions on the surface of CuO species [67]. The weak basic sites reduced with introduction of lanthanum into OMA, which can also be partly ascribed to the decrease of the S_{BET} from 254.4 m^2/g to 245.0 m^2/g . Interaction of La³⁺ ions with hydroxyl groups on the surface of the catalyst with formation of La-O-Al bonds is another probable explanation for the decrease of the number of the weak basic sites. For

Table 2Catalytic performance of OM-Cu_xLa_yAl₁₀₀ catalysts during upgrading ethanol to n-butanol.^[a]

Sample	Conv.(%)	Sel.(%)						Yield of higher alcohols	
		Acetaldehyde	diethyl ether	Butyraldehyde	Ethyl acetate	n-Butanol	C6-C8 alcohols ^[b]	Others ^[c]	
OM-Cu _{6.3} La _{2.6} Al ₁₀₀	45.0	4.3	0.4	5.4	6.6	51.9	20.7	10.7	32.7
OM-Cu ₄ La _{2.6} Al ₁₀₀	52.2	2.7	0.4	5.0	6.8	51.9	20.3	12.9	37.7
OM-Cu _{0.8} La _{2.6} Al ₁₀₀	31.9	5.4	1.3	4.5	6.1	55.6	18.2	8.9	23.5
OM-La _{2.6} Al ₁₀₀	4.1	0	47.8	0.3	5.2	46.7	0	0	1.9
OM-Cu ₄ Al ₁₀₀	45.9	4.6	4.3	4.9	6.6	48.8	20.0	10.8	31.6
OMA	9.9	0	76.1	0.4	2.6	20.9	0	0	2.1

[a] Conversion and selectivity are obtained at steady-state; reaction conditions: 533 K, 3 MPa (N₂), LHSV = 2 ml/(h·g_{cat}), N₂/ethanol(v/v) = 500:1; [b] C6-C8 alcohols include 2-ethylbutanol, 1-hexanol, 2-ethylhexanol and 1-octanol; [c] Other products mainly include 1,1-diethoxyethane, ethyl butyrate, butyl acetate, 2-ethylbutyraldehyde, hexaldehyde, etc.

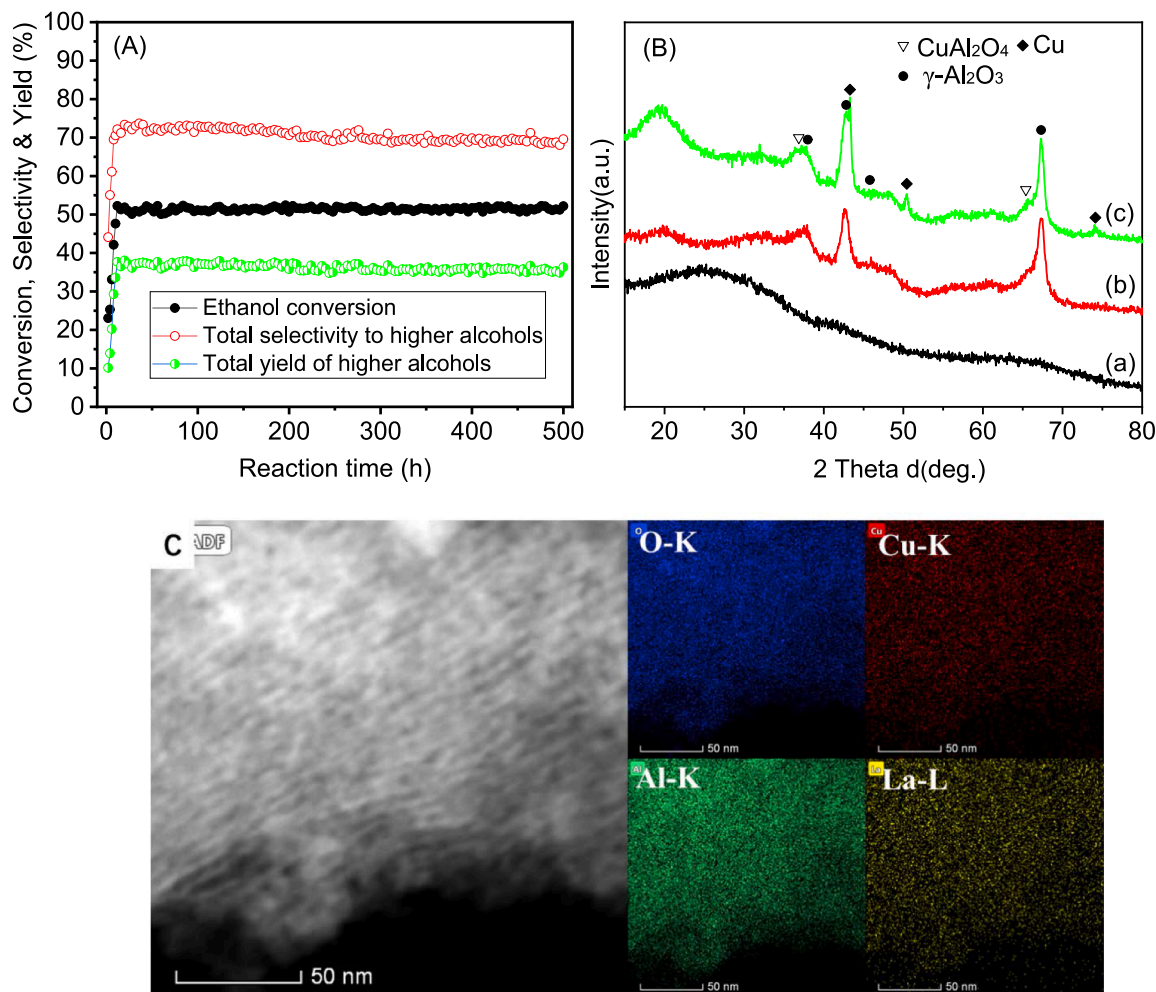


Fig. 10. (A) Long-term stability of the OM-Cu₄La_{2.6}Al₁₀₀ catalyst, (B) XRD patterns of the OM-Cu₄La_{2.6}Al₁₀₀ catalyst before (a) and after reaction of 12 h (b) and 500 h (c), and (C) HAADF-TEM images of OM-Cu₄La_{2.6}Al₁₀₀ catalyst after 500 h reaction and corresponding EDX mapping images.

the catalysts containing both copper and lanthanum, the number of weak basic sites increased with the increase of Cu RMN from 0.8 to 6.3, while numbers of medium and strong basic sites first increased and then decreased with the achievement of maximum at the Cu RMN of 4. As mentioned earlier, the increase of basic sites with increase of the Cu RMN can be explained by the reduction of coordinatively unsaturated Al³⁺ and interaction between La³⁺ ions and surface -OH [68].

3.6. Catalytic performance and long-term evaluation

Catalytic performance of this series of catalysts for upgrading of ethanol to higher alcohols was presented in Table 2. OMA exhibited very

high selectivity to dimethyl ether (76.1%) instead of to n-butanol (20.9%) with 9.9% conversion of ethanol, due to the existence of strong acid sites as confirmed by NH₃-TPD. Since introduction of lanthanum into OMA reduced the strength of the strong acid sites on the surface of Al₂O₃ substrate, the selectivity to dimethyl ether decreased to 47.8% and the selectivity to n-butanol increased to 46.7%, but with much lower conversion of ethanol (4.1%). The introduction of copper into OMA significantly improved the conversion of ethanol and selectivities to acetaldehyde and products synthesized via acetaldehyde intermediate including butyraldehyde, ethyl acetate, higher alcohols et al., indicating surface copper species are the main active component for ethanol dehydrogenation. Additionally, the high selectivity to higher alcohols

can be ascribed to the increase of the number of the weak basic sites and especially the presence of a small number of medium and strong basic sites on the surface of OM-Cu₄Al₁₀₀ catalyst [69]. For the series of catalysts containing lanthanum, ethanol conversion and the yield of higher alcohols significantly increased from 4.1% and 1.9% to 31.9% and 23.5%, respectively, with increase of the Cu RMN from 0 to 0.8, and then increase to 52.2% and 37.7% with increase of the Cu RMN to 4. Further increase of the Cu RMN to 6.3 resulted in a decrease in catalytic activity, which should be mainly attributed to partial collapse of the ordered structure. Thus, the OM-Cu₄La_{2.6}Al₁₀₀ catalyst exhibited the highest ethanol conversion of 52.2% with a yield to higher alcohols up to 37.7%. The best catalytic performance of OM-Cu₄La_{2.6}Al₁₀₀ catalyst should be related to the presence of a large amount of Cu_{Octa}²⁺ and acid-base sites with appropriate strength as shown in Table 1.

Regarding the industrial applications, a perfect catalyst is required to not only have excellent catalytic activity and selectivity to target products but also an outstanding stability. As Shown in Fig. 10(A), the ethanol conversion, total selectivity to and yield of higher alcohols were almost kept constant (51.4%, 70.6% and 36.3%, respectively) after a gradual activation in-situ of ca. 12 h during long-term evaluation of 500 h. XRD analysis for the OM-Cu₄La_{2.6}Al₁₀₀ catalyst before and after reaction of 12 h and 500 h were also shown in Fig. 10(B). After the initial gradual crystallization process in ethanol upgrading to higher alcohols, the structure of OM-Cu₄La_{2.6}Al₁₀₀ catalyst kept stable except for the appearance of a little Cu crystal phase. Especially, the diffraction peaks characteristic of CuAl₂O₄ for the OM-Cu₄La_{2.6}Al₁₀₀ catalyst after reaction of 500 h were still as weak and diffuse as those for the OM-Cu₄La_{2.6}Al₁₀₀ catalyst after reaction of 12 h, confirming the high stability of CuAl₂O₄ active phase. In addition, ordered mesoporous structures could be still found in the HAADF-TEM images of OM-Cu₄La_{2.6}Al₁₀₀ catalyst after 500 h reaction, while the EDX mapping images also showed the copper and lanthanum species were in high dispersion, which can further accountable for the excellent stability of the catalyst. The high thermal stability of the CuAl₂O₄ and ordered mesoporous structures were also responsible for the excellent stability of the catalyst. Simultaneously, the embedding effect of CuAl₂O₄ and La₂O₃ into ordered mesoporous framework further improved the stability of the OM-Cu₄-La_{2.6}-Al₁₀₀ catalyst. The construction of Cu-supported catalysts based on the CuAl₂O₄ and Cu_{Octa}²⁺ will provide a new way for the preparation of highly efficient and anti-sintering Cu-based catalysts.

3.7. Catalytic mechanism

The product distribution over OM-Cu_xLa_yAl₁₀₀ catalysts indicated that higher alcohols were produced from ethanol via a sequence of steps known as the Guerbet reaction. In this cascade process, ethanol first experienced dehydrogenation to form acetaldehyde, followed by aldol condensation of acetaldehyde and subsequently hydrogenation of crotonaldehyde to produce n-butanol, which might undergo further coupling with ethanol or n-butanol to give higher carbon number alcohols. Since introduction of copper into OMA and OM-La_{2.6}Al₁₀₀ dramatically improved the ethanol conversion and yield to higher alcohols, the highly dispersed CuAl₂O₄ on the OM-Cu_xLa_yAl₁₀₀ catalysts was main active phase for the ethanol dehydrogenation. Mendes et al. [70] found that Rh-Cu/Al₂O₃ catalyst with Cu_{Octa}²⁺ species showed high activity in n-hexanol dehydrogenation. The Cu_{0.5}Zn_{0.5}Fe₂O₄ catalyst prepared by microwave combustion showed isopropanol conversion of 59.8% in isopropanol dehydrogenation at 523 K due to existence of a large number of octahedrally coordinated Cu²⁺ and Fe³⁺ active sites [71]. Riittonen et al. [29] found that Cu/γ-Al₂O₃ catalyst with low loading of Cu exhibited much higher activity for ethanol dehydrogenation than the γ-Al₂O₃ support and then inferred the anti-spinel structure containing Cu_{Octa}²⁺ ions were active phase for ethanol dehydrogenation and then the formation of n-butanol. As confirmed by the results of XRD, TPR, XPS, UV-vis, and ²⁷Al NMR, Cu species on the surface of OM-

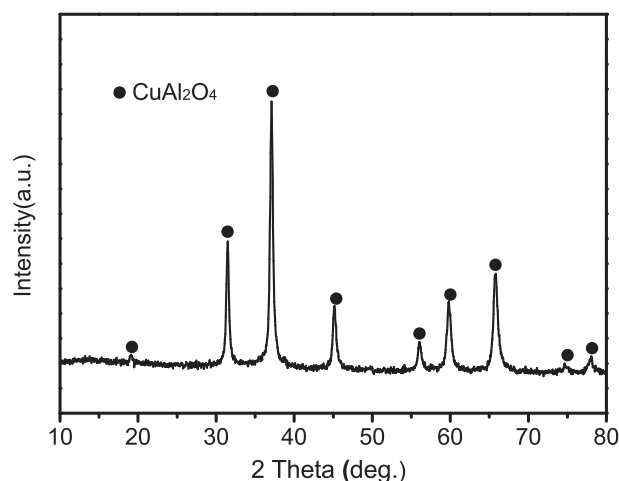


Fig. 11. XRD spectrum of CuAl₂O₄ made by our laboratory.

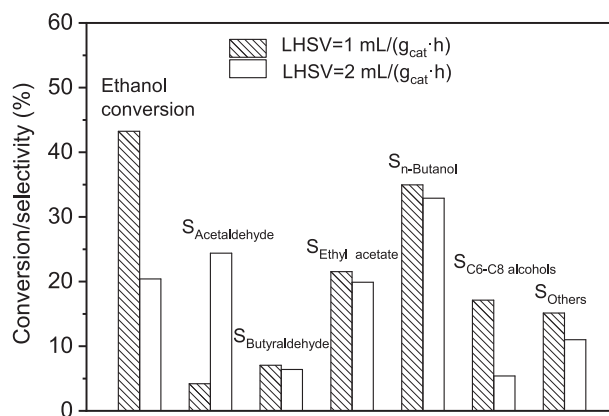


Fig. 12. Catalytic performance of stoichiometric CuAl₂O₄ for ethanol upgrading to higher alcohols at different LHSVs.

Cu_xLa_yAl₁₀₀ catalysts was mainly presented in the form of highly dispersed CuAl₂O₄ while the Cu_{Octa}²⁺ predominantly existed on the surface of CuAl₂O₄ crystalline grains. So, the surface Cu_{Octa}²⁺ species were expected to be exactly the active sites for ethanol dehydrogenation in ethanol upgrading to higher alcohols.

In order to further prove ethanol dehydrogenation capability of the octahedrally coordinated Cu²⁺, stoichiometric CuAl₂O₄ spinel was subsequently prepared as a catalyst for ethanol upgrading in this study. As shown in Fig. 11, the prepared sample exhibited standard diffraction patterns of CuAl₂O₄, indicating a successful synthesis of the CuAl₂O₄ catalyst. It can be seen from Fig. 12, the product distribution over CuAl₂O₄ catalyst was almost the same as that over OM-Cu_xLa_yAl₁₀₀ catalysts with acetaldehyde, ethyl acetate, n-butanol, 2-ethylbutanol, and 1-hexanol as main intermediates and products. Furthermore, the selectivity to acetaldehyde increased while the selectivities to ethyl acetate, n-butanol, 2-ethylbutanol, and 1-hexanol decreased with increasing the LHSV, suggesting acetaldehyde is probably the intermediate product for the formation of ethyl acetate, n-butanol, 2-ethylbutanol, and 1-hexanol. These results suggested the CuAl₂O₄ phase itself can be used as a catalyst for ethanol upgrading to higher alcohols vis Guerbet mechanism. In view of its low specific surface area (12.3 m²·g⁻¹) and large crystal size of CuAl₂O₄ (26 nm) due to high temperature calcination, the CuAl₂O₄ phase has high ethanol conversion activity in essence. It is known that stoichiometric CuAl₂O₄ spinel consists of 58% Cu_{Tetra}²⁺ sites and 42% Cu_{Octa}²⁺ sites with much higher fraction of Cu_{Octa}²⁺ sites on the surface of CuAl₂O₄ spinel [61]. Thus, the Cu_{Octa}²⁺ species were the exact active sites for ethanol dehydrogenation and hydrogenation of

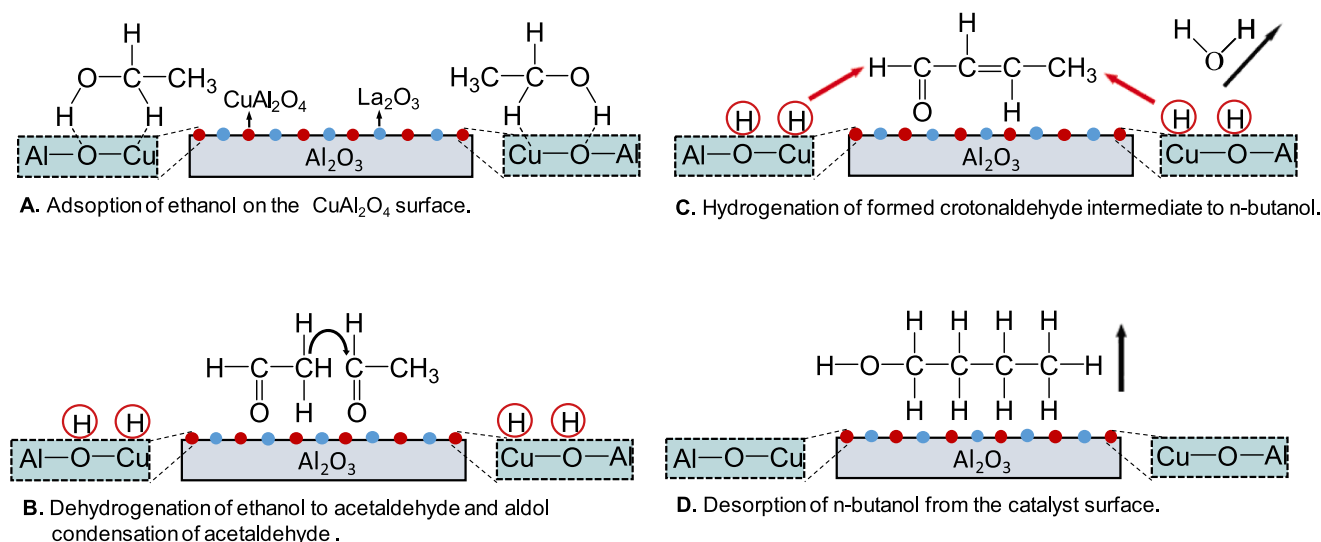


Fig. 13. Suggested reaction mechanism for catalytic upgrading of ethanol to n-butanol over $\text{OM-Cu}_x\text{La}_y\text{Al}_{100}$ catalysts.

crotonaldehyde in ethanol upgrading to higher alcohols. Furthermore, Lewis acid-base sites on the surface of CuAl_2O_4 crystalline grains might promote aldol condensation of acetaldehyde. Additionally, the high selectivities to ethyl acetate (21.5% and 19.9% at the different LHSVs) could be related to the accompanying surface $\text{Cu}^{2+}_{\text{tetra}}$ sites [29], which needs to be addressed more intensively in the future. The suggested reaction mechanism of ethanol upgrading over the $\text{OM-Cu}_x\text{La}_y\text{Al}_{100}$ catalysts was schematically depicted in Fig. 13. Specifically, ethanol molecule initially adsorbs on the octahedrally coordinated Cu^{2+} and nearby O^{2-} ions, and then α -hydrogen and hydroxyl hydrogen of ethanol are removed to form acetaldehyde. As confirmed by results of NH_3 -TPD and CO_2 -TPD, there were not only a large number of medium acid sites but also basic sites, especially medium and strong basic sites on the surface of $\text{OM-Cu}_x\text{La}_y\text{Al}_{100}$ catalyst. Then acetaldehyde condensation occurred at adjacent Lewis acid and medium/strong base sites to form 3-hydroxybutyraldehyde [69,72], which was followed by dehydration to crotonaldehyde and finally hydrogenation of crotonaldehyde on the $\text{Cu}^{2+}_{\text{Octa}}$ to n-butanol. It is obvious that the $\text{Cu}^{2+}_{\text{Octa}}$ and acid-base sites respectively promoted the different elementary step and ethanol was selectively converted to n-butanol via the four cascade reactions with the cooperation of $\text{Cu}^{2+}_{\text{Octa}}$ and acid-base sites. In general, the excellent catalytic activity and selectivity of Cu-La-Al composite metal oxides were ascribed to the synergy of a large amount of $\text{Cu}^{2+}_{\text{Octa}}$ and acid-base sites.

4. Conclusions

Ordered mesoporous Cu-La-Al composite oxides synthesized via an evaporation induced self-assembly method exhibited excellent catalytic activity, selectivity and stability in ethanol upgrading to higher alcohols. The highly dispersed CuAl_2O_4 phase rich in octahedrally coordinated Cu^{2+} was highly active for dehydrogenation/hydrogenation while a large number of acid and base sites mainly provided by alumina support and evenly spread La_2O_3 , respectively, were responsible for the aldol condensation. The ultra-high stability of the catalyst was largely attributed to presence of the new-type highly stable CuAl_2O_4 active phase besides the embedding effect of CuAl_2O_4 and La_2O_3 into skeleton of the composite oxides.

CRedit authorship contribution statement

Haojian Zhao: Writing – original draft, Data curation, Resources, Investigation. **Xinyue Shen:** Resources, Investigation. **Jinfeng He:**

Resources, Investigation. **Jianyao Kou:** Resources, Investigation. **Tingjie Tao:** Resources, Investigation. **Yu Can:** Resources, Investigation. **Hao Huang:** Resources, Investigation. **Dahao Jiang:** Writing – review & editing, Project administration, Supervision, Funding acquisition. **Lili Lin:** Resources, Investigation. **Xiaonian Li:** Resources, Investigation, Project administration.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dahao Jiang reports financial support was provided by Natural Science Foundation of Zhejiang Province. Dahao Jiang has patent Ordered mesoporous copper-rare earth metal-aluminum composite oxide catalysts and their preparation method and Application pending to Hangzhou Tianzheng Patent Brand Firm Co., Ltd.

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References

- [1] C. Angelici, B.M. Weckhuysen, P.C.A. Bruijninx, Chemocatalytic conversion of ethanol into butadiene and other bulk chemicals, *ChemSusChem* 6 (9) (2013) 1595–1614.
- [2] B.E. Ribeiro, Beyond commonplace biofuels: Social aspects of ethanol, *Energy Policy* 57 (2013) 355–362.
- [3] R.F. Britto, C.A. Martins, Emission analysis of a diesel engine operating in diesel-ethanol dual-fuel mode, *Fuel* 148 (2015) 191–201.
- [4] A. Atmanli, N. Yilmaz, A comparative analysis of n-butanol/diesel and 1-pentanol/diesel blends in a compression ignition engine, *Fuel* 234 (2018) 161–169.
- [5] K. Nanthagopal, B. Ashok, B. Saravanan, D. Patel, B. Sudarshan, R. Aaditya Ramasamy, An assessment on the effects of 1-pentanol and 1-butanol as additives with calophyllum inophyllum biodiesel, *Energy Convers. Manage.* 158 (2018) 70–80.
- [6] B. Ndaba, I. Chiyanzu, S. Marx, n-Butanol derived from biochemical and chemical routes: a review, *Biotechnol. Rep.* 8 (2015) 1–9.
- [7] J.T. Kozlowski, R.J. Davis, Heterogeneous catalysts for the Guerbet coupling of alcohols, *ACS Catal.* 3 (7) (2013) 1588–1600.
- [8] T.L. Jordison, C.T. Lira, D.J. Miller, Condensed-phase ethanol conversion to higher alcohols, *Ind. Eng. Chem. Res.* 54 (44) (2015) 10991–11000.
- [9] K. Koda, T. Matsui-ura, Y. Obora, Y. Ishii, Guerbet reaction of ethanol to n-butanol catalyzed by iridium complexes, *Chem. Lett.* 38 (8) (2009) 838–839.
- [10] G. Xu, T. Lammens, Q. Liu, X. Wang, L. Dong, A. Caiazzo, N. Ashraf, J. Guan, X. Mu, Direct self-condensation of bio-alcohols in the aqueous phase, *Green Chem.* 16 (8) (2014) 3971–3977.

- [11] R.L. Wingad, P.J. Gates, S.T.G. Street, D.F. Wass, Catalytic conversion of ethanol to n-butanol using ruthenium P-N ligand complexes, *ACS Catal.* 5 (10) (2015) 5822–5826.
- [12] K.-N. Tseng, S. Lin, J.W. Kampf, N.K. Szymczak, Upgrading ethanol to 1-butanol with a homogeneous air-stable ruthenium catalyst, *Chem. Commun.* 52 (14) (2016) 2901–2904.
- [13] Y. Xie, Y. Ben-David, L.J.W. Shimon, D. Milstein, Highly efficient process for production of biofuel from ethanol catalyzed by ruthenium pincer complexes, *J. Am. Chem. Soc.* 138 (29) (2016) 9077–9080.
- [14] S. Chakraborty, P.E. Piszal, C.E. Hayes, R.T. Baker, W.D. Jones, Highly selective formation of n-butanol from ethanol through the guerbet process: A tandem catalytic approach, *J. Am. Chem. Soc.* 137 (45) (2015) 14264–14267.
- [15] T.W. Birky, J.T. Kozlowski, R.J. Davis, Isotopic transient analysis of the ethanol coupling reaction over magnesia, *J. Catal.* 298 (2013) 130–137.
- [16] D.L. Carvalho, R.R. de Avillez, M.T. Rodrigues, L.E.P. Borges, L.G. Appel, Mg and Al mixed oxides and the synthesis of n-butanol from ethanol, *Appl. Catal. A* 415–416 (2012) 96–100.
- [17] I.-C. Marcu, D. Tichit, F. Fajula, N. Tanchoux, Catalytic valorization of bioethanol over Cu-Mg-Al mixed oxide catalysts, *Catal. Today* 147 (3–4) (2009) 231–238.
- [18] C. Yang, Z.Y. Meng, Bimolecular condensation of ethanol to 1-butanol catalyzed by alkali cation zeolites, *J. Catal.* 142 (1) (1993) 37–44.
- [19] T. Tsuchida, J. Kubo, T. Yoshioka, S. Sakuma, T. Takeguchi, W. Ueda, Reaction of ethanol over hydroxyapatite affected by Ca/P ratio of catalyst, *J. Catal.* 259 (2) (2008) 183–189.
- [20] S. Ogo, A. Onda, K. Yanagisawa, Selective synthesis of 1-butanol from ethanol over strontium phosphate hydroxyapatite catalysts, *Appl. Catal. A* 402 (1–2) (2011) 188–195.
- [21] J.H. Earley, R.A. Bourne, M.J. Watson, M. Poliakoff, Continuous catalytic upgrading of ethanol to n-butanol and >C₄ products over Cu/CeO₂ catalysts in supercritical CO₂, *Green Chem.* 17 (2015) 3018–3025.
- [22] D. Jiang, X. Wu, J. Mao, J. Ni, X. Li, Continuous catalytic upgrading of ethanol to n-butanol over Cu-CeO₂/AC catalysts, *Chem. Commun.* 52 (95) (2016) 13749–13752.
- [23] J. Pang, M. Zheng, L. He, L. Li, X. Pan, A. Wang, X. Wang, T. Zhang, Upgrading ethanol to n-butanol over highly dispersed Ni-MgAlO catalysts, *J. Catal.* 344 (2016) 184–193.
- [24] X.Y. Wu, G.Q. Fang, Z. Liang, W.H. Leng, K.Y. Xu, D.H. Jiang, J. Ni, X.N. Li, Catalytic upgrading of ethanol to n-butanol over M-CeO₂/AC (M = Cu, Fe, Co, Ni, Pd) catalysts, *Catal. Commun.* 100 (2017) 15–18.
- [25] D. Jiang, G. Fang, Y. Tong, X. Wu, Y. Wang, D. Hong, W. Leng, Z. Liang, P. Tu, L. Liu, K. Xu, J. Ni, X. Li, Multifunctional Pd@UiO-66 catalysts for continuous catalytic upgrading of ethanol to n-butanol, *ACS Catal.* 8 (12) (2018) 11973–11978.
- [26] J. Quesada, L. Faba, E. Díaz, S. Ordóñez, Enhancement of the 1-butanol productivity in the ethanol condensation catalyzed by noble metal nanoparticles supported on Mg-Al mixed oxide, *Appl. Catal. A* 563 (2018) 64–72.
- [27] X.Y. Wu, G.Q. Fang, Y.Q. Tong, D.H. Jiang, Z. Liang, W.H. Leng, L. Liu, P.X. Tu, H. J. Wang, J. Ni, X.N. Li, Catalytic upgrading of ethanol to n-butanol: Progress in catalyst development, *ChemSuschem* 11 (2018) 71–85.
- [28] T. Riittonen, E. Toukonen, D.K. Madhani, A.-R. Leino, K. Kordas, M. Szabo, A. Sapi, K. Arve, J. Wärnå, J.-P. Mikkola, One-pot liquid-phase catalytic conversion of ethanol to 1-butanol over aluminium oxide-The effect of the active metal on the selectivity, *Catalysts* 2 (1) (2012) 68–84.
- [29] T. Riittonen, K. Eränen, P. Mäki-Arvela, A. Shchukarev, A.-R. Rautio, K. Kordas, N. Kumar, T. Salmi, J.-P. Mikkola, Continuous liquid-phase valorization of bio-ethanol towards bio-butanol over metal modified alumina, *Renew. Energy* 74 (2015) 369–378.
- [30] A.V. Chistyakov, S.A. Nikolaev, P.A. Zharova, M.V. Tsodikov, F. Manenti, Linear α -alcohols production from supercritical ethanol over Cu/Al₂O₃ catalyst, *Energy* 166 (2019) 569–576.
- [31] Q. Zhang, J.J. Xu, F.Y. Fan, D.M. Sun, G.M. Xu, S. Zhang, Z.B. Zhu, Application of porous anodic alumina monolith catalyst in steam reforming of dimethyl ether: Cu/ γ -Al₂O₃/Al catalyst degradation behaviors and catalytic activity improvement by pre-competition impregnation method, *Fuel Process. Technol.* 119 (2014) 52–59.
- [32] Y. Wang, Y. Shen, Y. Zhao, J. Lv, S. Wang, X. Ma, Insight into the balancing effect of active Cu species for hydrogenation of carbon-oxygen bonds, *ACS Catal.* 5 (10) (2015) 6200–6208.
- [33] G.Y. Yergazyeva, K. Dossomov, M.M. Mambetova, P.Y. Strizhak, H. Kurokawa, B. Baizhomartov, Effect of Ni, La, and Ce Oxides on a Cu/Al₂O₃ catalyst with low copper loading for ethanol non-oxidative dehydrogenation, *Chem. Eng. Technol.* 44 (10) (2021) 1890–1899.
- [34] Y. Dai, B. Lim, Y. Yang, C.M. Cobley, W. Li, E.C. Cho, B. Grayson, P.T. Fanson, C. T. Campbell, Y. Sun, Y. Xia, A Sinter-resistant catalytic system based on platinum nanoparticles supported on TiO₂ nanofibers and covered by porous Silica, *Angew. Chem., Int. Ed.* 49 (44) (2010) 8165–8168.
- [35] S.H. Joo, J.Y. Park, C.-K. Tsung, Y. Yamada, P. Yang, G.A. Somorjai, Thermally stable Pt/mesoporous silica core-shell nanocatalysts for high-temperature reactions, *Nat. Mater.* 8 (2) (2009) 126–131.
- [36] P. Lu, C.T. Campbell, Y. Xia, A Sinter-resistant catalytic system fabricated by maneuvering the selectivity of SiO₂ deposition onto the TiO₂ surface versus the Pt nanoparticle surface, *Nano Lett.* 13 (10) (2013) 4957–4962.
- [37] Q. Liu, J. Gao, F. Gu, X. Lu, Y. Liu, H. Li, Z. Zhong, B. Liu, G. Xu, F. Su, One-pot synthesis of ordered mesoporous Ni-V-Al catalysts for CO methanation, *J. Catal.* 326 (2015) 127–138.
- [38] Y.S. Yun, D.S. Park, J. Yi, Effect of nickel on catalytic behaviour of bimetallic Cu-Ni catalyst supported on mesoporous alumina for the hydrogenolysis of glycerol to 1,2-propanediol, *Catal. Sci. Technol.* 4 (9) (2014) 3191–3202.
- [39] H. Arbag, S. Yasyerli, N. Yasyerli, G. Dogu, T. Dogu, I.G. Osojnik Črnivec, A. Pintar, Coke minimization during conversion of biogas to syngas by bimetallic tungsten-nickel incorporated mesoporous alumina synthesized by the one-pot route, *Ind. Eng. Chem. Res.* 54 (8) (2015) 2290–2301.
- [40] Q. Yuan, A.X. Yin, C. Luo, L.D. Sun, Y.W. Zhang, W.T. Duan, H.C. Liu, C.H. Yan, Facile synthesis for ordered mesoporous gamma-aluminas with high thermal stability, *J. Am. Chem. Soc.* 130 (2008) 3465–3472.
- [41] Y. Zang, X. Dong, C. Wang, One-pot synthesis of mesoporous Cu-SiO₂-Al₂O₃ bifunctional catalysts for hydrogen production by dimethyl ether steam reforming, *Chem. Eng. J.* 313 (2017) 1583–1592.
- [42] S.R. Yenumala, P. Kumar, S.K. Maity, D. Shee, Hydrodeoxygenation of karanja oil using ordered mesoporous nickel-alumina composite catalysts, *Catal. Today* 348 (2020) 45–54.
- [43] F. Liu, X. Zheng, J. Chen, Y. Zheng, L. Jiang, Controlling the synthesis and application of nanocrystalline spherical and ordered mesoporous alumina with high thermal stability, *RSC Adv.* 5 (114) (2015) 93917–93925.
- [44] R. Chein, Z. Yang, Experimental study on dry reforming of biogas for syngas production over Ni-based catalysts, *ACS Omega* 4 (25) (2019) 20911–20922.
- [45] M.F. Luo, P. Fang, M. He, Y.L. Xie, In situ XRD, Raman, and TPR studies of CuO/Al₂O₃ catalysts for CO oxidation, *J. Mol. Catal. A: Chem.* 239 (2005) 243–248.
- [46] K. Patel, V. Blair, J. Douglas, Q. Dai, Y. Liu, S. Ren, R. Brennan, Structural effects of lanthanide dopants on alumina, *Sci. Rep.* 7 (2017) 39946.
- [47] B. Yan, Y. Gao, B. Wang, X. Fan, F. Alam, J. Li, T. Jiang, Enhanced carbon dioxide oxidative dehydrogenation of 1-butene by iron-doped ordered mesoporous alumina, *ChemCatChem* 9 (24) (2017) 4480–4483.
- [48] Y.u. Xing, Z. Liu, S. Gomez, S.L. Suib, Tuning of texture and structure of copper-containing nanocomposite oxide materials, *J. Phys. Chem. C* 112 (5) (2008) 1446–1454.
- [49] V.K. Kaushik, C. Sivaraj, P.K. Rao, ESCA characterization of copper alumina catalysts prepared by deposition precipitation using urea hydrolysis, *Appl. Surf. Sci.* 51 (1991) 27–33.
- [50] L.F. Chen, P.J. Guo, M.H. Qiao, S.R. Yan, H.X. Li, W. Shen, H.L. Xu, K.N. Fan, Cu/SiO₂ catalysts prepared by the ammonia-evaporation method: Texture, structure, and catalytic performance in hydrogenation of dimethyl oxalate to ethylene glycol, *J. Catal.* 257 (2008) 172–180.
- [51] SeongUk Yun, V. Gulians, Novel bimetallic Cu-Pd nanoparticles as highly active low-temperature WGS catalysts, *Ind. Eng. Chem. Res.* 59 (5) (2020) 1789–1799.
- [52] A.M. Bahmanpour, B.P. Le Monnier, Y.-P. Du, F. Héroguel, J.S. Luterbacher, O. Kröcher, Increasing the activity of the Cu/CuAl₂O₄/Al₂O₃ catalyst for the RWGS through preserving the Cu²⁺ ions, *Chem. Commun.* 57 (9) (2021) 1153–1156.
- [53] Z.P. Huang, Z.P. Chen, Y.C. Chen, Y.Y. Hu, Synergistic effects in iron-copper bimetal doped mesoporous γ -Al₂O₃ for Fenton-like oxidation of 4-chlorophenol: Structure, composition, electrochemical behaviors and catalytic performance, *Chemosphere* 203 (2018) 442–449.
- [54] H. Reinhard, K.Z. Helmut, U. Hans-Peter, Surface properties and reduction behavior of calcined CuO/Al₂O₃ and CuO-NiO/Al₂O₃ catalysts, *J. Catal.* 69 (1981) 475–486.
- [55] K. Shimizu, H. Kawabata, H. Maeshima, A. Satsuma, T. Hattori, Intermediates in the Selective Reduction of NO by Propene over Cu-Al₂O₃ Catalysts: Transient in-Situ FTIR Study, *J. Phys. Chem. B* 104 (2000) 2885–2893.
- [56] K.-I. Shimizu, H. Maeshima, H. Yoshida, A. Satsuma, T. Hattori, Spectroscopic characterisation of Cu-Al₂O₃ catalysts for selective catalytic reduction of NO with propene, *Phys. Chem. Chem. Phys.* 2 (10) (2000) 2435–2439.
- [57] T. Yamamoto, T. Tanaka, S. Suzuki, R. Kuma, K. Teramura, Y. Kou, T. Funabiki, S. Yoshida, NO reduction with CO in the presence of O₂ over Cu/Al₂O₃ (3) – Structural analysis of active species by means of XAFS and UV/VIS/NIR spectroscopy, *Top. Catal.* 18 (2002) 113–118.
- [58] L.L. Pérez, S. Perdriau, G.T. Brink, B.J. Kooi, H.J. Heeres, I. Melián-Cabrera, Stabilization of self-assembled alumina mesophases, *Chem. Mater.* 25 (6) (2013) 848–855.
- [59] J.M.R. Gallo, C. Bisio, G. Gatti, L. Marchese, H.O. Pastore, Physicochemical characterization and surface acid properties of mesoporous [Al]-SBA-15 obtained by direct synthesis, *Langmuir* 26 (8) (2010) 5791–5800.
- [60] J.H. Kwak, J.Z. Hu, D.H. Kim, J. Szanyi, C.H.F. Peden, Penta-coordinated Al³⁺ ions as preferential nucleation sites for BaO on gamma-Al₂O₃: An ultra-high-magnetic field Al-27 MAS NMR study, *J. Catal.* 251 (2007) 189–194.
- [61] T. Tangcharoen, W. Klysubun, C. Kongmark, Synchrotron X-ray absorption spectroscopy and cation distribution studies of NiAl₂O₄, CuAl₂O₄, and ZnAl₂O₄ nanoparticles synthesized by sol-gel auto combustion method, *J. Mol. Struct.* 1182 (2019) 219–229.
- [62] P. Alphonse, B. Faure, Thermal stabilization of alumina modified by lanthanum, *Microporous Mesoporous Mater.* 196 (2014) 191–198.
- [63] C. Morterra, G. Magnacca, A case study: Surface chemistry and surface structure of catalytic aluminas, as studied by vibrational spectroscopy of adsorbed species, *Catal. Today* 27 (1996) 497–532.
- [64] H. Yi, X. Zhang, X. Tang, S. Zhao, C. Ma, W. Han, L. Song, Promotional effects of transition metal modification over Al₂O₃ for CH₃SH Catalytic Oxidation, *Chemistryselect* 4 (34) (2019) 9901–9907.
- [65] N.P. Parvaneh, D. Gérard, M.S. Mahdi, Activity of γ -Al₂O₃-based Mn, Cu, and Co oxide nanocatalysts for selective catalytic reduction of nitric oxide with ammonia, *Turk. J. Chem.* 41 (2017) 272–281.

- [66] N.K. Mishra, P. Kumar, .C. Srivastava V, U.L. Štangar, Synthesis of Cu-based catalysts for hydrogenolysis of glycerol to 1,2-propanediol with in-situ generated hydrogen, *J. Environ. Chem. Eng.* 9 (2021) 105263.
- [67] P. Kumar, R.C. Korošec, U.L. Štangar, Highly active and efficient Cu-based hydrotalcite-like structured materials as reusable heterogeneous catalysts used for transcarboxylation reaction, *J. Colloid Interface Sci.* 585 (2021) 549–559.
- [68] T. Yamamoto, T. Hatsui, T. Matsuyama, T. Tanaka, T. Funabiki, Structures and acid-base properties of La/Al₂O₃ role of La addition to enhance thermal stability of γ -Al₂O₃, *Chem. Mater.* 15 (25) (2003) 4830–4840.
- [69] J.I. Di Cosimo, C.R. Apesteguía, M.J.L. Ginés, E. Iglesia, Structural requirements and reaction pathways in condensation reactions of alcohols on Mg_yAlO_x catalysts, *J. Catal.* 190 (2) (2000) 261–275.
- [70] F.M.T. Mendes, M. Schmal, The cyclohexanol dehydrogenation on Rh CuAl₂O₃ catalysts Part 1. Characterization of the catalyst, *Appl. Catal., A* 151 (1997) 393–408.
- [71] M.H. Mahmoud, A.M. Hassan, A.-A. Said, H.H. Hamdeh, Structural; magnetic and catalytic properties of nanocrystalline Cu_{0.5}Zn_{0.5}Fe₂O₄ synthesized by microwave combustion and ball milling methods, *J. Mol. Struct.* 1114 (2016) 1–6.
- [72] J.I. Di Cosimo, V.K. Díez, M. Xu, E. Iglesia, C.R. Apesteguía, Structure and surface and catalytic properties of Mg-Al basic oxides, *J. Catal.* 178 (2) (1998) 499–510.